

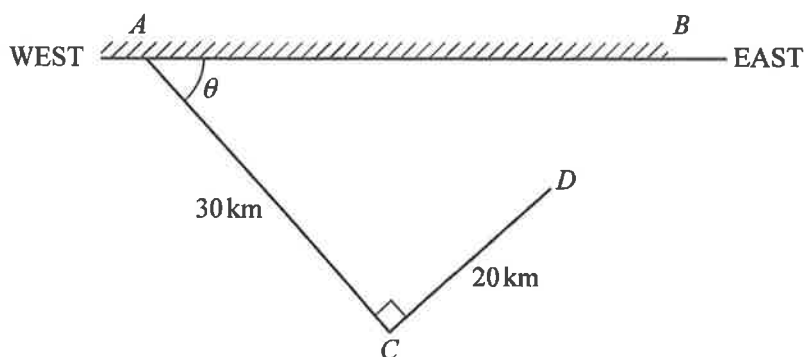
Basic

1. Solve the equation $3 \cos A = \sec A - 5 \tan A$ for $0^\circ < A < 360^\circ$. [7]
[N20/I/7]
2. (i) Prove that $\frac{\sin \theta}{1 - \cos \theta} - \frac{1}{\sin \theta} = \cot \theta$. [4]
- (ii) Hence solve the equation $\frac{\sin \theta}{1 - \cos \theta} - \frac{1}{\sin \theta} = 3 \tan \theta$ for $0^\circ \leq \theta \leq 180^\circ$. [3]
[N19/II/2]
3. Prove that $(\sec \theta - \tan \theta)^2 = \frac{1 - \sin \theta}{1 + \sin \theta}$. [4]
[N16/I/11(a)]

Intermediate

4. (a) Prove the identity $\tan 2\theta (2 \cos \theta - \sec \theta) = 2 \sin \theta$. [4]
- (b) Hence solve the equation $\tan 2\theta (2 \cos \theta - \sec \theta) = \frac{1}{2} \sec \theta$ for $0^\circ \leq \theta \leq 360^\circ$. [4]
[N22/II/3]
5. A calculator must not be used in this question.
 - (a) Show that $\tan 75^\circ = 2 + \sqrt{3}$. [4]
 - (b) Use the result from part (a) to find an expression for $\sec^2 75^\circ$, in the form $a + b\sqrt{3}$ where a and b are integers. [2]
 - (c) Hence express $\cos^2 75^\circ$ in the form $c + d\sqrt{3}$, where c and d are fractions. [2]
[N22/II/5]

6.



Two countries are separated by a border that runs from west to east. Villages A and B lie on the border. A helicopter flies 30 km from village A to a village C . Angle BAC is θ° . The helicopter then travels 20 km to a village D at an angle of 90° to AC as shown. You can assume that the helicopter always flies at the same height.

- (a) Show that the shortest distance of village D from the border is $(30 \sin \theta - 20 \cos \theta)$ km. [3]
- (b) Express $30 \sin \theta - 20 \cos \theta$ in the form $R \sin(\theta - \alpha)$ where $R > 0$ and $0^\circ \leq \alpha \leq 90^\circ$. [4]
- (c) Find the value of θ if village D is 10 km from the border. [2]

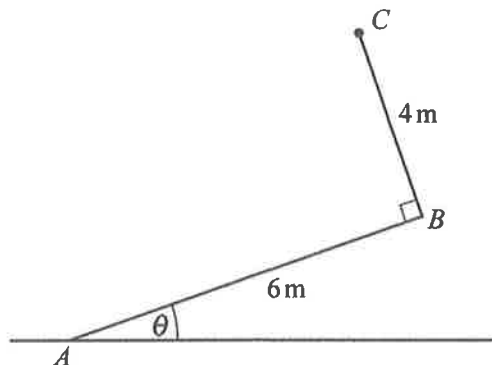
[N22/II/7]

7. (a) Prove the identity $\frac{\sin \theta}{1 - \cos \theta} - \frac{1}{\sin \theta} = \cot \theta$. [4]

(b) Hence solve the equation $\frac{\sin 2\theta}{1 - \cos 2\theta} - \frac{1}{\sin 2\theta} = -2$ for $0^\circ \leq \theta \leq 180^\circ$. [4]

[N21/I/10]

8.



The diagram shows two rods AB and BC rigidly joined at B so that $\angle ABC = 90^\circ$. The lengths of AB and BC are 6 m and 4 m respectively. A light is positioned at C . The point A is fixed on horizontal ground and the rod AB rotates in a vertical plane with the rod AB inclined at an angle θ to the ground.

(a) Show that the height, h metres, of C above the ground is given by

$$h = 6 \sin \theta + 4 \cos \theta. \quad [2]$$

(b) Express h in the form $R \sin(\theta + \alpha)$, where $R > 0$ and $0^\circ \leq \alpha \leq 90^\circ$. [4]

(c) Find the value of h and the corresponding value of θ if the light at C is to be positioned as high as possible. [3]

[N21/II/4]

9. The expression $7 \cos \theta + 4 \sin \theta$ is defined for $0 \leq \theta \leq \pi$ radians.

(i) Using $R \cos(\theta - \alpha)$, where $R > 0$ and $0 < \alpha < \frac{\pi}{2}$, solve the equation $7 \cos \theta + 4 \sin \theta = 6$. [4]

(ii) State the largest and smallest values of $80 - (7 \cos \theta + 4 \sin \theta)^2$ and find the corresponding values of θ . [4]

[N20/II/1]

10. A , B and C are the angles of a triangle.

(i) Show that $\tan C = -\tan(A + B)$. [2]

Angle $A = 45^\circ$ and angle $B = 60^\circ$.

(ii) Without using a calculator, find $\tan C$ in the form $a + \sqrt{b}$, where a and b are integers. [3]

[N18/I/2]

11. (i) Show that $8 \sin^2 x + 2 \cos^2 x$ can be written as $a + b \cos 2x$, where a and b are integers. [2]

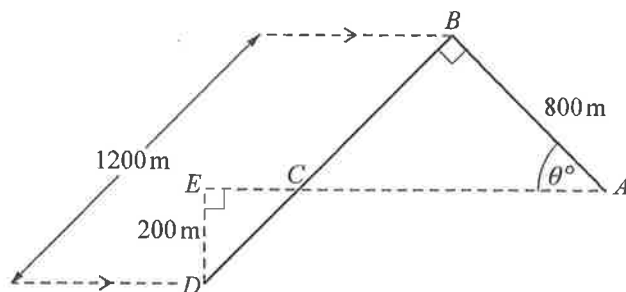
Hence

(ii) state the period and amplitude of $8 \sin^2 x + 2 \cos^2 x$, [2]

(iii) sketch the graph of $y = 8 \sin^2 x + 2 \cos^2 x$ for $0 \leq x \leq 2\pi$ radians. [3]

[N18/I/8]

12.



The diagram above represents the path of a skier. He takes the ski-lift from A to B , a journey of 800 m along a slope inclined at θ° to the horizontal. From B he skis 1200 m downhill to finally reach D after passing through the point C . The points E , C and A are at the same level and E is 200 m directly above D .

- (i) Given that angle ABC is 90° , express BC in terms of $\tan \theta$ and hence obtain an expression for CD . [2]
- (ii) Hence show that DE can be expressed in the form $a \cos \theta + b \sin \theta$, where a and b are constants. [3]
- (iii) By expressing DE in the form $R \cos(\theta + \alpha)$, find the value of θ . [5]

[N18/II/5]

13. (i) Show that $\frac{\sec x + \operatorname{cosec} x}{\sec x - \operatorname{cosec} x} = \frac{\tan x + 1}{\tan x - 1}$. [3]

- (ii) Hence find, for $0 \leq x \leq 2\pi$, the values of x in radians for which $\frac{\sec x + \operatorname{cosec} x}{\sec x - \operatorname{cosec} x} = \frac{5}{2}$. [3]

[N17/I/5]

14. It is given that $y_1 = 2 \sin x + 1$ and $y_2 = -\cos 2x$.

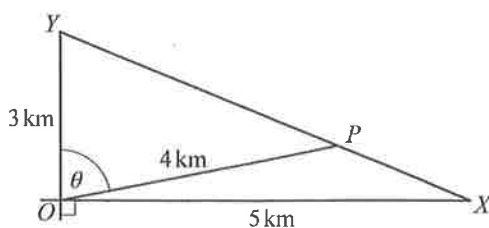
- (i) State the amplitude and the period, in degrees, of (a) y_1 , (b) y_2 . [2]

For the interval $0^\circ \leq x \leq 360^\circ$,

- (ii) solve the equation $y_1 = y_2$, [4]
- (iii) sketch, on the same diagram, the graphs of y_1 and y_2 , [4]
- (iv) find the set of values of x for which $y_1 - y_2 > 0$. [2]

[N17/II/10]

15.



The diagram shows an area of rough ground bordered by a straight road XY . The point O is such that $OX = 5$ km, $OY = 3$ km and angle $XOY = 90^\circ$. A cross-country runner leaves O and reaches P on the road XY after running for 4 km in a straight line inclined at an angle θ to OY .

- (i) Express the shortest distance of P from OX and from OY in terms of θ . [1]
- (ii) Show that $20 \cos \theta + 12 \sin \theta = 15$. [3]
- (iii) Express $20 \cos \theta + 12 \sin \theta$ in the form $R \cos(\theta - \alpha)$, where $R > 0$ and α is acute. [4]
- (iv) Find the value of θ . [2]

[N17/II/11]

16. (i) Using $\cos 3x = \cos(2x + x)$, show that $\cos 3x$ may be expressed as $\cos x(1 - 4 \sin^2 x)$. [3]
- (ii) Find all the values of x between 0° and 360° for which $2 \cos 3x = 15 \sin x \cos x$. [5]

[N16/II/3]

17. Given that $f(x) = 4 \cos^2 x - 2 \sin^2 x$,

- (i) express $f(x)$ in the form $a \cos 2x + b$, stating the value of each of the integers a and b , [3]
- (ii) state the greatest and least values of $f(x)$, [2]
- (iii) state the period and amplitude of $f(x)$. [2]

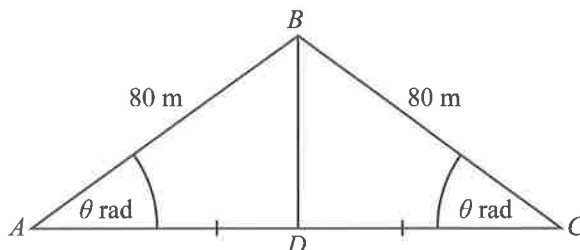
[N15/I/6]

18. (i) Show that $4 \tan \theta + 2 \cot \theta = 5 \sec \theta$ can be expressed as $2 \sin^2 \theta - 5 \sin \theta + 2 = 0$. [3]

- (ii) Hence solve the equation $4 \tan 2x + 2 \cot 2x = 5 \sec 2x$ for $0^\circ < x < 360^\circ$. [4]

[N15/I/8]

19.



A farmer fences part of his land. He puts fences around the perimeter of the triangular field ABC and also from B to D , where D is the mid-point of AC . Angle $BAC =$ angle $BCA = \theta$ radians and the lengths of AB and BC are 80 m.

- (i) Show that L m, the length of fencing needed, can be expressed in the form $p + q \sin \theta + r \cos \theta$, where p , q and r are constants to be found. [3]
- (ii) Express L in the form $p + R \sin (\theta + \alpha)$, where $R > 0$ and α is an acute angle. [4]
- (iii) Given that the farmer uses exactly 310 m of fencing, find the value of θ . [3]

[N15/II/9]

Advanced

20. Do not use a calculator in this question.

- (i) Show that $\frac{\sin (\alpha + \beta) + \sin (\alpha - \beta)}{\cos (\alpha + \beta) + \cos (\alpha - \beta)} = \tan \alpha$. [2]

Angles A and B are both acute and $\sin (A - B) = -\frac{3}{5}$.

- (ii) What can be deduced about the size of B relative to A ? [1]
- (iii) Find the value of $\cos (A - B)$. [2]
- (iv) Given that $\cos (A + B) = \frac{7}{25}$ and $\tan (A + B) = \frac{24}{7}$, state the value of $\sin (A + B)$. [1]
- (v) Using the identity from **part (i)**, show that $\tan A = \frac{1}{3}$. [2]
- (vi) Hence find the exact value of $\tan B$. [3]

[N20/I/12]